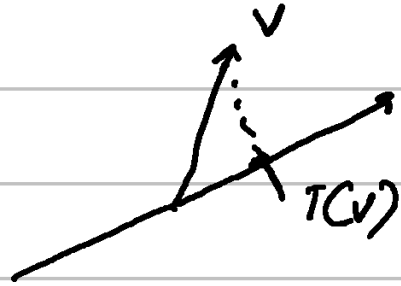


Lecture 30

Ex. 1 Projection

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$



Linear transformation is also mapping.

→ A transformation is linear when →

$$T(u+v) = T(u) + T(v)$$

$$T(cv) = c T(v)$$

$$\Rightarrow T(cu + dw) = cT(u) + dT(w)$$

ex2 T : shift the whole plane by v_0

$$v \Rightarrow v + v_0$$

$$T(u + w) = u + w + v_0$$

$$T(u + w) = T(u) + T(w) = u + w + 2v_0$$

Not a linear transformation.

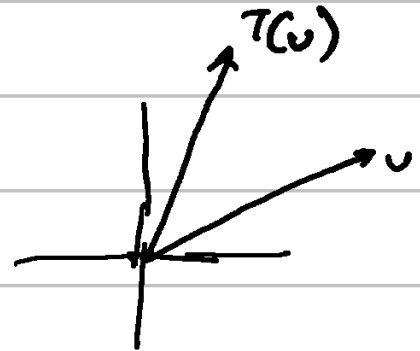
Fact $\rightarrow T(0) = 0$

The above statement must be true for a transformation to be linear.

$$\underline{\text{Proof}} \rightarrow T(3 \cdot v_0) = 3 \cdot \overset{\rightarrow 0}{T(v)} = 0$$

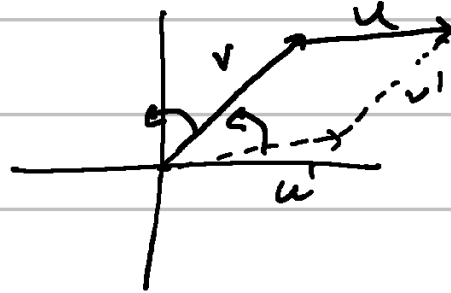
$$\text{Hence } T(0) = 0$$

ex 3 T : Rotation by 45°
 $\mathbb{R}^2 \rightarrow \mathbb{R}^2$



$T(c \cdot v) = c \cdot T(v)$ [Rotation doesn't change length]

$T(u+v) = T(u) + T(v)$



Rotation matrix = $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

eg

$$T(v) = \underbrace{Av}$$

always a linear transformation

eg

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$T(v) = Av \rightarrow 3 \times 1$$

$$Av = 2 \times 1$$

$$A \rightarrow 2 \times 3$$

→ How much information is needed to know $T(v)$ for all inputs.

$T(v_1), T(v_2)$

If we know these two transformations we know all their linear combinations as well.

⇒ So, let's say the transformation is from $\mathbb{R}^n \rightarrow \mathbb{R}^n$

Then knowing $T(v_i)$ for n independent vectors would give us the transformation for all vectors.

$T(v_1), T(v_2), \dots, T(v_n) \} \rightarrow$ complete transformation.

$v_i \rightarrow$ Basis for space of input

Coordinates come from a basis!

eg
$$v = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$$

where $\{v_i\}$ is basis for vector space, then basically c_i are respective co-ordinates.

$\Rightarrow$ Conventionally, $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, these vectors are taken as basis then co-ordinates become simple.

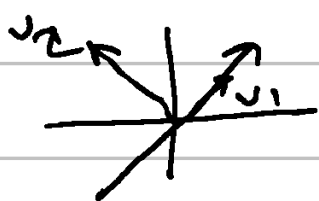
It is called the standard basis.

- Construct a matrix A that represents a linear transformation T .

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

- ① Choose basis $v_1, \dots, v_n$ for inputs
- ② Choose basis $w_1, \dots, w_m$ for outputs.

eg For projection transformations, if we take the input and output basis to be along the line & $\perp$ to line



$$v = c_1 v_1 + c_2 v_2 \quad \text{along } \perp$$

$$T(v) = c_1 v_1$$

So, the matrix that kills the perpendicular component is $\rightarrow$

$$T(v) = A v$$

↓

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

⇒ We can also see that, the bases are eigenvectors of projection matrix and the matrix comes out to be diagonal matrix of λ 's.

⇒ Now to construct A from the given bases for inputs and outputs.

⇒ finding A from

$v_1, \dots, v_n$

$w_1, \dots, w_m$

⇒ 1st column of $A \Rightarrow T(v_1) = \boxed{a_{11}}w_1 + a_{21}w_2 + \dots$

→ The coordinates of transformation of input basis are the respective columns of A !!

$$A \begin{pmatrix} \text{input} \\ \text{coord} \end{pmatrix} = \begin{pmatrix} \text{output} \\ \text{coords} \end{pmatrix}$$

eg $T = \frac{d}{dx}$

Input : $c_1 + c_2 x + c_3 x^2$

input basis : $1, x, x^2$

output : $1, x \Rightarrow c_2 + 2c_3 x$

Taking a derivative is a linear transformation.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \rightarrow \begin{bmatrix} c_2 \\ 2c_3 \end{bmatrix}$$

2×3